

A two-channel instability region for the spinorial CKN inequality

Candidate partial result for arXiv:2504.16909

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Status. Candidate partial result; verifier verdict: *likely valid*. The theorem below proves that mixing channels genuinely enlarges the instability region. It does not determine the optimal region over all mixed channels.

1 Source problem

The source is J. Dolbeault, M. J. Esteban, R. L. Frank, and M. Loss, *The CKN inequality for spinors: symmetry and symmetry breaking*, arXiv:2504.16909v2 [1]. In Remark 26 on PDF page 30, after deriving single-channel instability regions, the authors write:

“It is not unlikely that by mixing several channels one might increase the region of instability. Finding the optimal region is an open problem.”

The complete source paragraph, the adjacent theorem, and the added-in-proof novelty warning are reproduced below.

$$p > p_5(\alpha) := \frac{1}{2\alpha - 1} \quad \text{and} \quad \alpha > \frac{1}{6} \approx 0.788675.$$

Remark 26. Note that we have obtained symmetry breaking by destabilization of the symmetric spinor $\phi_\star$ along single channel functions $\varphi \in \mathcal{S}_k^m$. For such functions, it is easily seen that the above choices of k and m are the best possible ones. It is not unlikely that by mixing several channels one might increase the region of instability². Finding the optimal region is an open problem.

Theorem 27. Let $\alpha \in \mathbb{R} \setminus \Lambda$ and $p \in (2, 6)$. Symmetric optimizers are linearly unstable and $C_{\alpha,p} < C_{\alpha,p}^\star$ if one of the following conditions is satisfied:

- (i) $\alpha \in (-1/2, (1 - \sqrt{3})/4)$ and $p > \min\{p_1(\alpha), p_2(\alpha)\}$,
- (ii) $\alpha \in ((3 + \sqrt{3})/6, 1)$ and $p > p_5(\alpha)$,
- (iii) $\alpha > 1$.

Moreover, $C_{\alpha,p} < C_{\alpha,p}^\star$ if $\alpha < -1/2$ and $(-(1 + \alpha), p)$ is in an unstable region (i), (ii) or (iii).

See Fig. 4 for a plot of p_1 and p_2 in Case (i). The functions $\alpha \mapsto p_j(\alpha)$ with $j = 1, 2, 5$ are strictly monotone on the intervals considered in Theorem 27, and their inverses $p \mapsto \alpha_j(p)$ are the functions which appear in Theorem 6.

²Added in proof : Recent work of M.J. Esteban and R. Frank suggests that this is the case.

2 Setup

We use the source's logarithmic-cylinder formulation on $\mathbb{R} \times \mathbb{S}^2$, where surface measure $d\omega$ on $\mathbb{S}^2$ is normalized to be a probability measure. For $\alpha > -1/2$, the symmetric critical point is

$$\phi_\star(s, \omega) = u_\star(s)\chi_0, \quad \chi_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The source computes its real Hessian $\mathcal{Q}$. We only need the following specialization of that formula. Let $\psi \in L^2(\mathbb{S}^2, \mathbb{C}^2)$ satisfy

$$\|\psi\|_{L^2(\mathbb{S}^2)} = 1, \quad \int_{\mathbb{S}^2} \langle \chi_0, \psi \rangle d\omega = 0,$$

and let $w \in H^1(\mathbb{R})$ be real. Put

$$\delta(\psi) := \int_{\mathbb{S}^2} |\operatorname{Re} \langle \chi_0, \psi \rangle|^2 d\omega.$$

Then the source Hessian gives

$$\mathcal{Q}[w\psi] = \int_{\mathbb{R}} \left\{ |w'|^2 + \mu(\psi)|w|^2 - A(1 + (p-2)\delta(\psi)) \frac{|w|^2}{\cosh^2(Bs)} \right\} ds, \quad (1)$$

where

$$A = \frac{p}{2} \left(\frac{1}{2} - \alpha \right)^2, \quad B = \frac{p-2}{2} \left(\frac{1}{2} - \alpha \right), \quad \mu(\psi) = \left\| \left(\sigma \cdot L + \frac{1}{2} - \alpha \right) \psi \right\|_2^2. \quad (2)$$

The zero-mean hypothesis makes the final positive rank-one term in the full Hessian vanish.

The scalar minimization used repeatedly in the source is

$$\inf_{\|w\|_2=1} \int_{\mathbb{R}} \left(|w'|^2 - C \frac{|w|^2}{\cosh^2(Bs)} \right) ds = -\frac{1}{4} \left(\sqrt{4C + B^2} - B \right)^2. \quad (3)$$

This is the ground-state formula for the Pöschl–Teller operator.

3 Proof intuition

The two lowest channels used separately in the source have angular labels $(k, m) = (1, 1/2)$ and $(-2, 1/2)$. Their upper components are positive multiples of the same real harmonic Y_1^0 , while their lower components are opposite multiples of Y_1^1 . There is therefore a unique normalized combination in which the lower component cancels. The result is the spinor $(\sqrt{4\pi}Y_1^0, 0)^T$. Its real upper component carries the entire norm, so the destabilizing nonlinear factor is maximal: $\delta = 1$. The cost of this gain is only the weighted average of the two angular energies. Once the angular calculation is made, the radial minimization remains the same scalar Pöschl–Teller problem as in the source and yields an explicit threshold.

4 The two-channel theorem

Theorem 1 (Explicit mixed-channel instability). *Let*

$$-\frac{1}{2} < \alpha < 0, \quad 2 < p < 6, \quad d := \frac{1}{2} - \alpha.$$

If

$$p > p_{\text{mix}}(\alpha) := 2\sqrt{1 + \frac{2}{d^2}}, \quad (4)$$

then the symmetric critical point is linearly unstable. Consequently,

$$C_{\alpha,p} < C_{\alpha,p}^*.$$

The interval $(p_{\text{mix}}(\alpha), 6)$ is nonempty for every $\alpha \in (-1/2, 0)$. In particular, for

$$\frac{1 - \sqrt{3}}{4} \leq \alpha < 0$$

this gives a nonempty instability region while Theorem 27 of the source gives none from any single channel. Hence channel mixing genuinely increases the published instability region.

Proof. Write $Z_j := \sqrt{4\pi} Y_1^j$. Appendix B of the source gives

$$\chi_1^{1/2} = \begin{pmatrix} \sqrt{2/3} Z_0 \\ \sqrt{1/3} Z_1 \end{pmatrix}, \quad \chi_{-2}^{1/2} = \begin{pmatrix} \sqrt{1/3} Z_0 \\ -\sqrt{2/3} Z_1 \end{pmatrix}.$$

Define the normalized mixed channel

$$\psi := \sqrt{\frac{2}{3}} \chi_1^{1/2} + \sqrt{\frac{1}{3}} \chi_{-2}^{1/2}. \quad (5)$$

The displayed formulas show the exact cancellation

$$\psi = \begin{pmatrix} Z_0 \\ 0 \end{pmatrix}.$$

Since Y_1^0 is real, normalized in ordinary surface measure, and has mean zero, while the source uses probability surface measure, we have

$$\|Z_0\|_{L^2(d\omega)} = 1, \quad \int_{\mathbb{S}^2} Z_0 d\omega = 0, \quad \delta(\psi) = \int_{\mathbb{S}^2} Z_0^2 d\omega = 1.$$

The two constituent channels are orthonormal eigenvectors of $\sigma \cdot L$, with eigenvalues 1 and -2 . Therefore

$$\begin{aligned} \mu(\psi) &= \frac{2}{3} \left(\frac{3}{2} - \alpha \right)^2 + \frac{1}{3} \left(-\frac{3}{2} - \alpha \right)^2 \\ &= \alpha^2 - \alpha + \frac{9}{4} = d^2 + 2. \end{aligned}$$

Insert these identities into (1). The potential coefficient is

$$C = A(1 + (p-2)\delta) = A(p-1) = \frac{p(p-1)}{2} d^2, \quad B = \frac{p-2}{2} d.$$

After optimizing w by (3), the least Hessian value in this one-radial-profile, two-angular-channel subspace is

$$\begin{aligned} E_{\text{mix}}(\alpha, p) &= d^2 + 2 - \frac{1}{4} \left(\sqrt{4C + B^2} - B \right)^2 \\ &= d^2 + 2 - \frac{d^2}{16} \left(\sqrt{9p^2 - 12p + 4} - (p - 2) \right)^2. \end{aligned} \quad (6)$$

For $p > 2$, the function

$$R(p) := \sqrt{9p^2 - 12p + 4} - (p - 2)$$

is strictly increasing. Indeed,

$$R'(p) = \frac{9p - 6}{\sqrt{9p^2 - 12p + 4}} - 1 > 0,$$

where the last inequality follows by squaring the positive quantities. Equation (6) is zero exactly when

$$R(p) = 4\sqrt{1 + \frac{2}{d^2}}.$$

Squaring once and simplifying gives the unique positive solution

$$p = 2\sqrt{1 + \frac{2}{d^2}}.$$

Thus (4) is equivalent to $E_{\text{mix}} < 0$. The Hessian criterion from the source now implies linear instability and $C_{\alpha, p} < C_{\alpha, p}^*$.

Finally, $p_{\text{mix}}(\alpha) < 6$ is equivalent to $d > 1/2$, which is exactly $\alpha < 0$. The comparison with the source follows from its Theorem 27, whose negative- α single-channel condition requires $\alpha < (1 - \sqrt{3})/4$. $\square$

Corollary 2 (Reflected range). *Using the source symmetry $\alpha \mapsto -(1 + \alpha)$, symmetric optimizers are linearly unstable for*

$$-1 < \alpha < -\frac{1}{2}, \quad p > 2\sqrt{1 + \frac{2}{(\alpha + 3/2)^2}}.$$

Proof. Set $\tilde{\alpha} := -(1 + \alpha) \in (-1/2, 0)$ and apply Theorem 1. The source's reflection identity transfers the strict inequality of constants back to α , and $1/2 - \tilde{\alpha} = \alpha + 3/2$. $\square$

5 Why this is not yet the optimal region

The construction above uses a common radial function in the two channels. Allowing independent real profiles w_+ and w_- produces the matrix Schrödinger operator

$$-\frac{d^2}{ds^2}I + \begin{pmatrix} (3/2 - \alpha)^2 & 0 \\ 0 & (-3/2 - \alpha)^2 \end{pmatrix} - \frac{A}{\cosh^2(Bs)} \left[I + (p-2)vv^\top \right], \quad v = \begin{pmatrix} \sqrt{2/3} \\ \sqrt{1/3} \end{pmatrix}. \quad (7)$$

For $\alpha \neq 0$, the constant diagonal matrix and the rank-one matrix $vv^\top$ do not commute. A constant channel rotation therefore cannot reduce (7) to scalar Pöschl–Teller operators. The unrestricted Hessian also contains analogous higher-angular blocks. The packet proves a complete theorem for the explicit subspace (5), but does not exclude lower eigenvalues in these larger blocks and hence does not solve the source’s optimal-region problem.

6 Verification and numerical sanity check

The companion script `code/check_two_channel_instability.py` checks:

- the exact upper/lower component cancellation;
- the threshold identity at six values of α ;
- four points where the mixed energy is negative but both constituent one-channel energies are positive;
- one direct 200,001-point quadrature of the explicit Pöschl–Teller ground-state Rayleigh quotient.

At $(\alpha, p) = (-0.10, 5.50)$ the values are approximately

$$E_{\text{mix}} = -0.3625, \quad E_{(1,1/2)} = 0.69423, \quad E_{(-2,1/2)} = 0.89738.$$

The quadrature and closed mixed energy agree to better than $2 \cdot 10^{-9}$. These computations are sanity checks and are not used in the proof.

7 Novelty status and review recommendation

A bounded search on 9 August 2026 covered the run’s lightweight indexes and parsed arXiv corpus, together with arXiv searches for the exact source title, the authors Esteban and Frank, “mixing several channels,” and spinorial CKN instability. No separate public paper containing threshold (4) was found. Novelty confidence is nevertheless only moderate: version 2 of the source, posted 13 July 2026, adds a footnote saying that recent work of Esteban and Frank suggests that mixing does increase the region. A human should check the two cancellation lines and the threshold algebra, and should contact the source authors before any external originality claim.

References

- [1] J. Dolbeault, M. J. Esteban, R. L. Frank, and M. Loss, *The CKN inequality for spinors: symmetry and symmetry breaking*, arXiv:2504.16909v2, 2026.